

Smart Attendance System

Next-Gen Dynamic QR & Geofenced Campus Attendance Platform

Executive Summary: The Smart Attendance System is a cloud-native, mobile-first campus application engineered to permanently eliminate proxy attendance ('buddy punching'), paper sign-in sheets, and manual roll calls. It combines **time-expiring rotating QR codes (HMAC-SHA256)**, **GPS geofencing (Haversine equation)**, and **device fingerprinting** to guarantee that only physically present students in the lecture hall can mark their attendance.

1. The Core Problem vs. The Smart Solution

Traditional classroom attendance suffers from three major flaws that compromise institutional integrity:

- **Proxy Attendance (Buddy Punching):** Friends mark attendance for absentees using roll calls or static sign-in sheets.
- **QR Code Forwarding:** In basic digital apps, students photograph the QR code and send it via WhatsApp to friends staying in dorms.
- **Classroom Disruption:** Manual roll calls consume 10 to 15 minutes of every lecture, wasting thousands of teaching hours per semester.

How Attendly Solves It: With Attendly, taking attendance takes under **30 seconds** for an entire auditorium. The professor projects a rotating QR code on screen. Each code is mathematically valid for only **10 seconds**. Simultaneously, the student's mobile GPS coordinates are checked against the classroom's coordinates in real time.

2. How the System Works in Real Life (5 Simple Steps)

Step 1: Session Launch	The Teacher logs in, selects their Course/Class, and starts an Attendance Session. The system captures the teacher's classroom GPS coordinates and sets an allowable radius (e.g., 50 meters).
Step 2: Dynamic QR Generation	The teacher's screen displays a dynamic QR code containing a cryptographically signed HMAC token and timestamp. This QR automatically refreshes every 10 seconds. Screenshots sent to friends expire instantly.
Step 3: Instant 1-Tap Camera Scan	The Student opens the app, taps 'Scan QR' (which instantly launches the camera), and scans the code. The mobile browser simultaneously reads the student's high-precision device GPS.
Step 4: Real-time Cloud Verification	The server verifies: (1) Is the token authentic and unexpired? (2) Is the student's GPS within the teacher's allowed radius? (3) Has this student already marked? If valid, status is recorded in milliseconds!
Step 5: Live Analytics & Class View	The Teacher's dashboard instantly increments the live head-count. Teachers and admins can view granular attendance records, export CSVs, and audit absences.

3. Geolocation & Distance Calculation (How it Works)

A critical innovation in Attendly is its resilient **Haversine Great-Circle Distance Engine**. Earlier systems showed inaccurate distance errors when devices were side-by-side due to missing coordinate validation. Here is how precision is guaranteed:

- **High-Accuracy HTML5 Geolocation API:** Requests GPS hardware coordinates with `enableHighAccuracy: true` and a strict 10-second timeout.
- **Haversine Trigonometric Formulation:** Calculates exact curvature distance in meters between teacher (lat1, lon1) and student (lat2, lon2):

```
d = 2 * R * arcsin( sqrt( sin^2(delta_lat/2) + cos(lat1) * cos(lat2) * sin^2(delta_lon/2) ) )
where R = 6,371,000 meters (Earth's radius). Accurately calculates classroom proximity down to 1-2 meters.
```

- **Real-Time Feedback:** If a student scans from outside the allowed radius (e.g., dormitory or cafeteria), the scanner immediately displays a clear amber warning with exact measured distance vs. allowed boundary.

4. Technical Stack & Architecture

The system is built on a modern, ultra-fast serverless architecture designed for sub-100ms response times:

Layer	Technology	Role & Key Capability
Frontend Framework	Next.js 16 (App Router + Turbopack)	High performance React Server Components, fast hydration, pre-rendered static routes.
Styling & UI	Tailwind CSS + Lucide Icons	Glassmorphic modern campus theme, fully responsive for smartphones, tablets, and desktops.
Backend / Server	Next.js Edge / Node.js Route Handlers	51 optimized REST API routes with zero-cold-start latency and in-memory token hashing.
Database	Supabase (PostgreSQL 15)	Row-Level Security (RLS), real-time change data capture, automated timestamps and foreign keys.
Authentication	Supabase Auth + NextAuth Tokens	Role-Based Access Control (Admin, Teacher, Student) with secure HTTP-only session cookies.
Camera & QR Engine	HTML5 QrCode Scanner + QRCode.react	Instant hardware camera access with automatic fallback to environment-facing camera.
Hosting & Cloud	Vercel Edge Network + AWS Pooler	Global CDN caching, automatic CI/CD deployment on every Git push, 99.99% uptime.

5. User Roles & Key Features

Student Role (/student)

- 1-Tap QR Scanner directly accessing device camera with zero delay.
- Visual attendance history breakdown (% present, absent, excused).
- Course-by-course analytics with warnings for courses near attendance threshold (e.g. < 75%).
- GPS proximity checks providing transparent distance feedback.

Teacher Role (/teacher)

- One-click Session Creator with customizable expiration and geofence radius.
- Dynamic QR Projector with live student scan ticker and real-time head-count.
- Granular Class Roster Page showing each student's attendance history and attendance rates.
- Manual override capabilities (mark Present/Absent/Late) with justification notes.

Admin Role (/admin)

- Complete institutional overview: Total departments, courses, teachers, and enrolled students.
- Enrollment management and teacher course assignments.
- System audit logging: tracks logins, attendance overrides, and administrative actions.
- Comprehensive reporting tools with CSV export for academic records.

6. Ready-to-Use Demo Accounts (Pre-Seeded)

All accounts are pre-configured in the Supabase database with unified simple passwords for instant testing:

Role	Email	Password	Primary Screen
Admin	admin@attendly.edu	password123	/admin (System Dashboard)
Teacher	teacher@attendly.edu	password123	/teacher (Session & QR Hub)
Student 1	student@attendly.edu	password123	/student (Attendance Scanner)
Student 2	student2@attendly.edu	password123	/student (Attendance Scanner)
Student 3	student3@attendly.edu	password123	/student (Attendance Scanner)

Summary for Stakeholders: The Smart Attendance System delivers an airtight, tamper-proof, and frictionless attendance experience. By replacing traditional call-outs with rotating cryptographic QR tokens and geofencing, it protects institutional academic compliance while saving valuable teaching hours.